

Department of Computer Science & Engineering, SDMCET, Dharwad - 2

Course Title and code : Object Oriented Programming Laboratory (22UCSL403)

Semester / Division : IV / Both A and B division

Course Instructors : Prof. Indira R U and Prof. Chaitali Chate

<u>List of Programs for Lab Internals</u>	
1)	A vehicle is identified by how many wheels it has. A car, bike and truck are identified by their engine number, chassis number and manufacturer. Car <u>is-a</u> vehicle, Bike <u>is-a</u> vehicle and Truck <u>is-a</u> vehicle. Using this information, write a Java Program that displays all the information of vehicle, car, bike and truck. Also draw the class diagram for the problem scenario.
2)	Write a Java Program that takes a string as command-line argument and checks whether the string is palindrome or not. If the string is palindrome, then print proper message and also print its length; otherwise, generate a user-defined exception StringNotPalindromeException. If string is empty, generate another exception StringCannotBeEmpty.
3)	A chemical company named “ABC Chemicals Pvt. Ltd.” manufactures various chemicals. To automate and monitor the manufacturing process, they are planning to install an “ Automated Manufacturing and Maintenance System ”. One important task of this system is to sense the rise in temperature of the furnace and monitor its temperature. If the temperature of the furnace rises above 1500K, then it should immediately generate an alarm. It should also generate a separate alarm if the temperature of the furnace falls below 1000K. You are hired as a Java Programmer to develop this system. Write a Java Program which simulates the above scenario using exception handling mechanism.
4)	Write a Java program that creates three threads. The first thread prints prime numbers from 1 to 100; second thread prints prime numbers from 101 to 200 and third thread prints prime numbers from 201 to 300. Note: The three threads must call the same method generatePrime() to print the prime numbers.
5)	Write a Java program that simulates client-server interaction using threads. Methodology: Two threads should be created; one thread should act as server, and the other one should act as client. The server thread should accept a string from the client thread. Upon receiving and displaying the string, the server thread should send back the acknowledgement message “ Message Received ” to the client thread.

- 6) Student in SDMCET is identified by USN, first name, last name, branch and division. Write a Java program using generics to provide facility to sort the students based on USN, first name and division. The sort option is to be read by the user.
Note: The program must implement compareTo() method available in Comparable<T> interface to sort the students based on USN.
- 7) Write a Java program that provides a generic sort method, which can sort elements of any reference type.
- 8) Write a Java program to implement the following scenario:
Consider the following table available in the MySQL instance of EmployeeDB:
- | EmployeeInfo | |
|--------------|--------------|
| EmpID | Int |
| Name | varchar (40) |
| Department | varchar (30) |
| Salary | double |
- Write a Java program that displays contents of the above-mentioned table on the console.
- 9) Postgraduate course of the CSE department can accommodate a maximum of 20 students. Each PG student in the department is identified by his/her Roll No, USN, Name, Semester and Mobile Number. Using appropriate Collection class, write a Java Program to simulate the following scenarios:
- i. The initial capacity is 15.
 - ii. Only 5 students are enrolled for the course initially.
 - iii. Two students bearing Roll Numbers 3 and 5 voluntarily unroll from the course.
 - iv. After a month, two more students get enrolled in the course.
 - v. Print the information of all the students currently enrolled in the course.
- 10) Write a Java program to implement the following scenario:
Consider the following table available in the MySQL instance of LibraryDB:
- | BookInfo | |
|----------|--------------|
| ISBN | int |
| Title | varchar (40) |
| Author | varchar (30) |
| Price | double |
- Write a Java program that displays contents of the above-mentioned table on the console.
- 11) Bank Account Withdrawal Exception Scenario:
A banking application requires that a user cannot withdraw an amount greater than their current balance. Additionally, a minimum balance of Rs.1000 must be maintained.
Task: Create a custom checked exception InsufficientFundsException.
Implement withdraw (double amount) method in a BankAccount class. Throw the exception with

	a customized message if the withdrawal violates either condition (negative balance or dropping below the minimum). Write a main class to catch this exception and display a user-friendly message.																														
12)	A traffic system uses threads to simulate light changes. A "Controller" thread manages three light threads: Red, Yellow, and Green. Task: Create a program where the Red light thread stays active for 5 seconds, followed by Yellow for 2 seconds, and then Green for 5 seconds. Use a shared object and notifyAll() to trigger the next light thread once the previous one's sleep timer expires. Demonstrate the use of isAlive() to check which light is currently "on" before switching.																														
13)	Implement Producer-Consumer of single instance using synchronization and inter-thread communication.																														
14)	Model a User superclass with subclasses Doctor and Patient. Task: Include common attributes like ID and Name in the superclass. Add unique methods like prescribeMedicine() for Doctor and requestAppointment() for Patient. Use the super keyword to initialize common data in subclass constructors.																														
15)	Showcase the following using a Java application: i. There are 4 variables – default, public, private and protected. ii. These variables visibility is to be shown using the following table: <table><tr><th></th><th>Private</th><th>No Modifier</th><th>Protected</th><th>Public</th></tr><tr><td>Same class</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>Same package subclass</td><td>No</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>Same package non-subclass</td><td>No</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>Different package subclass</td><td>No</td><td>No</td><td>Yes</td><td>Yes</td></tr><tr><td>Different package non-subclass</td><td>No</td><td>No</td><td>No</td><td>Yes</td></tr></table>		Private	No Modifier	Protected	Public	Same class	Yes	Yes	Yes	Yes	Same package subclass	No	Yes	Yes	Yes	Same package non-subclass	No	Yes	Yes	Yes	Different package subclass	No	No	Yes	Yes	Different package non-subclass	No	No	No	Yes
	Private	No Modifier	Protected	Public																											
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Same package subclass	No	Yes	Yes	Yes																											
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Different package non-subclass	No	No	No	Yes																											
16)	Define a Flyable interface and an abstract class Animal. Create a class Bird that extends Animal and implements Flyable.																														